Lecture 28.03: Entropy as an Impurity Metric

📌 Topic Overview

To automate feature selection, a Decision Tree needs a mathematical formula to quantify impurity and disorder. That formula is **Entropy**. In this lecture, we study the theoretical formula, its range, and the exact step-by-step blackboard calculation.

📐 Mathematical Definition of Entropy

Entropy originated in physics (thermodynamics) and information theory (Claude Shannon). In Machine Learning:

> **Entropy measures the impurity, disorder, or uncertainty in a given dataset S.**

The formula for a dataset with c classes is:

$$E(S) = -\sum_{i=1}^{c} p_i \log_2(p_i)$$

Where:

- p_i is the proportion (probability) of class i in sample set S.
- $\log_2$ is logarithm base 2.

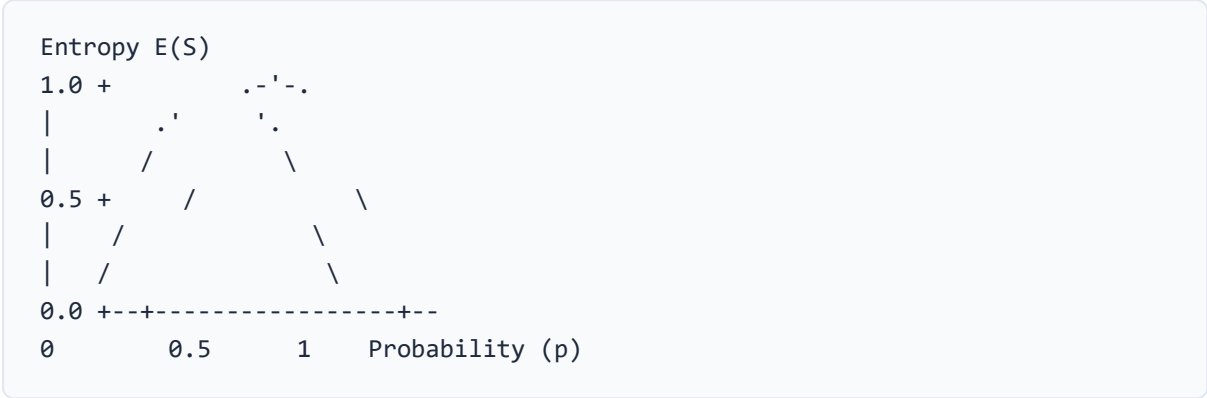
For a binary classification task (classes `Yes` and `No`):

$$E(S) = -p_{extyes} \log_2(p_{extyes}) - p_{extno} \log_2(p_{extno})$$

📈 Range and Inverted Parabola Curve

For binary classification:

- **Minimum Value = 0**: When the node is 100% pure (all `Yes` or all `No`). There is zero uncertainty!
- **Maximum Value = 1**: When there is a 50-50 split ($p_{extyes} = 0.5$, $p_{extno} = 0.5$). There is maximum uncertainty!



At $p = 0$ and $p = 1$, entropy drops to 0. At $p = 0.5$, entropy peaks at 1.0.

🎞️ Step-by-Step Numerical Example from Video

In the video, the instructor works out the following calculation on the blackboard:

Given Node:

$S = [3y, 3n]$ (3 Yes and 3 No, total samples = 6).

$$p_{ ext{yes}} = rac{3}{6} = rac{1}{2} = 0.5$$

$$p_{ ext{no}} = rac{3}{6} = rac{1}{2} = 0.5$$

Applying the formula:

$$E(S) = - \left(rac{1}{2} \cdot \log_2 rac{1}{2} ight) - \left(rac{1}{2} \cdot \log_2 rac{1}{2} ight)$$

Recall that $\log_2(0.5) = \log_2(2^{-1}) = -1$:

$$E(S) = - \left(rac{1}{2} \cdot (-1) ight) - \left(rac{1}{2} \cdot (-1) ight) = rac{1}{2} + rac{1}{2} = 1.0$$

Completely Pure Node:

For a node $S_{extpure} = [0y, 3n]$ (all 3 samples are No):

- $p_{extyes} = 0, p_{extno} = 1$.
- By convention, $0 \cdot \log_2(0) = 0$.
- $E(S_{extpure}) = -0 - (1 \cdot \log_2 1) = 0 - 0 = 0.0$.

💻 Python Implementation

```
import numpy as np

def calculate_entropy(yes_count, no_count):
    """
    Calculates Shannon Entropy for a binary classification node.
    Formula: E(S) = - [ p_yes * log2(p_yes) + p_no * log2(p_no) ]

    Returns:
        float: Entropy value in range [0.0 (Pure), 1.0 (Maximum Impurity)]
    """
    total = yes_count + no_count
    if total == 0:
        return 0.0

    p_yes = yes_count / total
    p_no = no_count / total

    # 0 * log2(0) is mathematically defined as 0
    term_yes = p_yes * np.log2(p_yes) if p_yes > 0 else 0.0
    term_no = p_no * np.log2(p_no) if p_no > 0 else 0.0

    entropy = -(term_yes + term_no)
    return entropy

# --- Function Call & Verification ---
print("Entropy [3 Yes, 3 No]:", calculate_entropy(3, 3))  # 1.0 (50-50 mix)
print("Entropy [0 Yes, 3 No]:", calculate_entropy(0, 3))  # 0.0 (Pure node)
print("Entropy [2 Yes, 4 No]:", round(calculate_entropy(2, 4), 3))  # 0.918
```

🔑 Key Takeaways

1. Entropy measures disorder or unpredictability.
2. For binary classification, Entropy is bounded between 0 (pure) and 1 (maximum uncertainty).

3. The lower the entropy of a node, the better and cleaner that subset is.